Amazingly, physicists had to search for seven years before they discovered a practical method by which nature might extract a hole's spin energy and put it to use. Their search led the physicists through one crazy method after another, all of which would work in principle but none of which showed much practical promise, before they finally discovered nature's cleverness. In Chapter 9 I shall describe this search and discovery, and its payoff: a black-hole "machine" for powering quasars and gigantic jets.

If, as we have seen, electric charge produces electric field lines that stick radially out of a hole's horizon, and spin produces a swirl in space around the hole, a distortion of the horizon's shape, and a storage of energy, then what happens when a hole has both charge and spin? Unfortunately, the answer is not terribly interesting; it contains little new. The hole's charge produces the usual electric field lines. The hole's spin creates the usual swirl of the hole's space, it stores the usual rotational energy, and it makes the horizon's equator bulge out in the usual manner. The only things new are a few rather uninteresting magnetic field lines, created by the swirl of space as it flows through the electric field. (These field lines are *not* a new form of "hair" on the hole; they are merely a manifestation of the interaction of the old, standard forms of hair: the interaction of the spin-induced swirl with the charge-induced electric field.) All the properties of a spinning, charged black hole are embodied in an elegant solution to the Einstein field equation derived in 1965 by Ted Newman at the University of Pittsburgh and a bevy of his students: Eugene Couch, K. Chinnapared, Albert Exton, A. Prakash, and Robert Torrence.

Not only can black holes spin; they can also pulsate. Their pulsations, however, were not discovered mathematically until nearly a decade after their spin; the discovery was impeded by a powerful mental block.

For three years (1969–71) John Wheeler's progeny "watched" black holes pulsate, and didn't know what they were seeing. The progeny were Richard Price (my student, and thus Wheeler's intellectual grandson), C. V. Vishveshwara and Lester Edelstein (students of Charles Misner's at the University of Maryland, and thus also Wheeler's intellectual grandsons), and Frank Zerilli (Wheeler's own student at Princeton). Vishveshwara,